

Knowledge Distillation

- transfer knowledge from a teacher model to a student model
 - student learns to match teacher outputs or intermediate features
- why used
 - compression: smaller model with similar accuracy
 - speed: faster inference (mobile / real-time)
 - regularization: teacher soft targets can improve generalization
- common objectives
 - logit distillation (soft targets)
 - $L = \text{CE}(y, p_s) + \lambda T^2, \text{KL}(p_t^{(T)} | p_s^{(T)})$
 - $p^{(T)} = \text{softmax}(z/T)$
 - feature distillation
 - match hidden representations (e.g., L2 / cosine loss)
- key knobs
 - temperature T (softness of targets) 초반에는 낮게 → 후반 갈수록 높게
 - weight λ (balance with supervised loss) softening sharpening

DINO

label 필요 X teacher → student

- self-supervised distillation for vision transformers (teacher-student)
 - student learns to predict teacher's output distribution
 - no labels required
 - core idea
 - two augmented views of the same image
 - teacher sees global crops, student sees global + local crops multi-crops
 - teacher targets are sharpened (low entropy) with temperature scheduling local → receptive field ↓
일부분 보고 global 정보 유추
- 모델이 local-to-global correspondence 학습

- objective
 - minimize cross-entropy between teacher and student outputs
 - encourages consistent representations across views
 - why it works
 - teacher is an **EMA** of the student (stable target) weight update → stable ↑ teacher가 자기 자신의 EMA [CLS] token 에 대해 나오는 output 맞춰가도록
 - **centering + temperature prevent collapse** (trivial constant outputs) ↪ 자기자신 평균 빼기 → 특정 output에 편중되는 현상 sharpening 공백 방지
 - outcome 바다속 평균
 - learns strong visual features that transfer well to downstream tasks
- collapse: 데이터의 의미 구분 학습 x → loss만 줄이도록 요행

DINO v2, v3 data curation data clustering graph anchoring (segmentation)

ControlNet

- add conditioning control to a pretrained diffusion model without destroying its generative ability
 - conditions: edge, depth, pose, segmentation, scribble, etc.
- core idea
 - copy the U-Net and attach a trainable control branch
 - keep the original pretrained U-Net frozen (or mostly frozen)
 - inject control features into the main U-Net via residual additions
- how it works (high-level)
 - control input c is encoded into multi-scale features
 - at each U-Net block, add control residuals to the main activations
 - "zero-conv" (initialized near zero) makes training stable
 - starts from "no control effect" and gradually learns control strength
- benefits
 - strong controllability with minimal degradation of text alignment and realism
 - works as a plug-in for many conditions
- limitations

- extra compute and memory (another branch)
- need paired data (image + condition) or condition extraction pipeline

T2I Adapter

- lightweight conditioning adapter for pretrained text-to-image diffusion
 - similar use cases: edges, depth, pose, segmentation, style hints
 - core idea
 - add a small adapter network that encodes the condition into feature maps
 - inject those features into the pretrained U-Net (often via residual connections)
 - typically fewer parameters than ControlNet
 - pretrained diffusion backbone stays frozen
 - how it works (high-level)
 - condition encoder produces multi-scale representations
 - features are fused into U-Net blocks (shallow additions)
 - train only the adapter (and sometimes small projection layers)
 - benefits
 - parameter-efficient, faster to train, lower memory overhead
 - easier to deploy when compute is tight
 - limitations
 - usually weaker controllability than full ControlNet
 - capacity may be insufficient for complex spatial constraints
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Neural Radiance Field (NeRF)

- continuous neural representation of a 3D scene
 - represents a scene as a function parameterized by a neural network